52.4: Interactions between organisms and the environment limit the distribution of species

Broadly speaking, species distributions are a consequence of both ecological factors and evolutionary history. Consider kangaroos, which are found in Australia and nowhere else in the world. Fossil evidence indicates that kangaroos and their close relatives originated in Australia, roughly 5 million years ago. By that time, Australia had moved close to its present location (by continental drift; see Concept 25.4), and it was not connected to other landmasses. Thus, kangaroos occur only in Australia in part because of an accident of history: The kangaroo lineage originated there at a point in time when the continent was geographically isolated.

But ecological factors are also important. To date, kangaroos have not dispersed (on their own) to other continents; hence, they are restricted to the continent on which they originated. And within Australia, kangaroos are found in some habitats but not in others. The red kangaroo, for example, occurs in the arid grasslands of central Australia, but not in the tall, open forests of eastern Australia. Moreover, kangaroos are not unusual in this respect—all species are found in some habitats but not others. Hence, ecologists ask not only *where* species occur, but also *why* species occur where they do: What ecological factors—biotic and abiotic—determine their distribution?

In many cases, both biotic and abiotic factors affect the distribution of a species, as is true of the saguaro cactus (*Carnegiea gigantea*). Saguaros are found almost exclusively in the Sonoran Desert of the southwestern United States and northwestern Mexico **(Figure 52.17)**. To the north, their range is limited by an abiotic factor: temperature. Saguaros tolerate freezing temperatures only briefly, typically for less than a day, and generally cannot survive at temperatures below −4°C(25°F). For the same reason, saguaros are rarely found at elevations above 1,200 m (4,000 feet).

Figure 52.17

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Distribution of the saguaro cactus in North America.
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Map of the extent of the Sonoran desert and observed Saguaro cacti. Saguaro cacti are observed throughout the Sonoran deserts of Arizona and Sonora.

Freezing temperatures strongly limit where saguaros are found, but other abiotic and biotic factors are also important.

Map of Southern California, Baja California, Baja California Sur, Arizona (borders both Southern California to the west and Mexico to the south), and Sonora (Mexico). The Sonoran Desert extends into southeastern California, southwest Arizona and northwest Sonora, and large portions of Baja California and Baja California Sur (excluding northwest Baja California and southeast Baja California Sur). Saguaro cacti are observed throughout the Sonoran deserts of Arizona and Sonora, but do not extend into Baja or Baja California Sur. Only a few occurrences of Saguaro cacti are observed in the Sonoran desert of far southeastern California.

However, temperature alone does not fully explain the distribution of saguaros, including why they are missing from the western portion of the Sonoran Desert. Water availability is critical because seedling survival typically requires consecutive years of moist conditions, something that may occur only a few times each century. Biotic factors almost certainly influence their distribution as well. Mice and grazers such as goats eat the seedlings, and bats pollinate the large, white flowers that open at night. Saguaros are also vulnerable to a deadly bacterial disease. Thus, for the saguaro, as for most other species, ecologists need to consider multiple factors and alternative hypotheses when attempting to explain the distribution of a species.

To see how ecologists might arrive at such an explanation, let's examine the ecological factors highlighted by the questions in the flowchart in **Figure 52.18**.

Figure 52.18

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Flowchart of factors limiting geographic distribution.
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A flowchart for why is species X absent from an area.

An ecologist studying factors limiting a species' distribution might consider questions like these. As suggested by the arrows leading from the "Yes" responses, the ecologist would answer all of these questions because more than one factor can limit a species' distribution.

Does dispersal limit its distribution? If yes, then area inaccessible or insufficient time. If no, do biotic factors (other species) limits its distribution. If yes, predation, parasitism, competition, disease. If no, do abiotic factors limit its distribution? If chemical factors, then water, oxygen, salinity, p H, soil nutrients, et cetera. If physical factors, then temperature, light, soil structure, fire, moisture, et cetera.

How might the importance of various abiotic factors differ for aquatic and terrestrial ecosystems?

*For suggested answer, see*Appendix A*.*

Dispersal and Distribution

One factor that contributes greatly to the global distribution of organisms is **dispersal**, the movement of individuals or gametes away from their area of origin or from centers of high population density. For example, while land-bound kangaroos have not reached Africa under their own power, other organisms that disperse more readily, such as some birds, have. The dispersal of organisms is critical to understanding the role of geographic isolation in evolution (see Concept 24.2) as well as the patterns of species distribution we see today, including that of the Pacific diatom discussed earlier in this chapter.

Natural Range Expansions and Adaptive Radiation

The importance of dispersal is most evident when organisms reach an area where they did not exist previously, called a *range expansion*. For instance, 200 years ago, the cattle egret (*Bubulcus ibis*) was found only in Africa and southwestern Europe. But in the late 1800s, some of these birds managed to cross the Atlantic Ocean and colonize northeastern South America. From there, cattle egrets gradually spread southward and also northward through Central America and into North America, reaching Florida by 1960 **(Figure 52.19)**. Today they have breeding populations as far west as the Pacific coast of the United States and as far north as southern Canada.

Figure 52.19

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Dispersal of the cattle egret in the Americas.
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Map of the Americas with dates of cattle egret spread to different regions, from first appearance in South America in 1937 to present distribution in North America.

Native to Africa, cattle egrets were first reported in South America in 1877. The color-coding and dates show the regions to which the birds had spread at different points in time.

Data are given in the following table:

Date	Specific Location	General Region
1937	Guyana, Surinam, French Guiana	Northeast South America
1943	Venezuela	Northeast South America
1951	Colombia	North South America
1956	Ecuador/Peru	Northwestern South America

1958	Panama, Costa Rica, Nicaragua, Honduras, El Salvador	Central America
1960	Florida, Gulf Coast	South Eastern North America
1961	South Eastern Mexico	Northern Central America
1965	North Western Mexico (extending into United States)	Northern Central America
1966	California, Baja California	North America
1970	Central United States	North America
1970	Chile	Southwestern South America
Current	Northern United States	North America

In rare cases, such long-distance dispersal can lead to adaptive radiation, the rapid evolution of an ancestral species into new species that fill many ecological niches. The diversity of Hawaiian silverswords is one example of adaptive radiation that was possible only with the long-distance dispersal of an ancestral tarweed from North America (see Figure 25.22).

Natural range expansions clearly show the influence of dispersal on distribution. However, opportunities to observe such dispersal directly are rare, so ecologists often turn to experimental methods to better understand the role of dispersal in limiting the distribution of species.

Species Transplants

To determine if dispersal is a key factor limiting the distribution of a species, ecologists may observe the results of intentional or accidental transplants of the species to areas where it was previously absent. For a transplant to be successful, some of the organisms must not only survive in the new area but also reproduce there sustainably. If a transplant is successful, then we can conclude that the *potential* range of the species is larger than its *actual* range; in other words, the species *could* live in certain areas where it currently does not.

In some cases, species introduced to new geographic locations have disrupted the communities and ecosystems to which they have been introduced (see Concept 56.1). Consequently, ecologists rarely move species to new regions. Instead, they document the outcome when a species reaches a new region in other ways, such as when a predator is introduced to control a pest species or when a species has been moved accidentally to the area.

Biotic Factors

Our next question is whether biotic factors—other species—limit the distribution of a species. Often, the ability of a species to survive and reproduce is reduced by its interactions with other species, such as predators (organisms that kill their prey) or herbivores (organisms that eat plants or algae). **Figure 52.20** shows how an herbivore, the long-spined sea urchin (*Centrostephanus rodgersii*), has affected the distribution of a food species. In certain marine ecosystems, there is often an inverse relationship between the abundance of sea urchins and seaweeds (multicellular algae, such as kelp). Where urchins that graze on seaweeds and other algae are common, large stands of seaweeds do not become established. As shown in Figure 52.20, Australian researchers tested whether *C. rodgersii* is a biotic factor limiting seaweed distribution. When this urchin was removed from experimental plots, seaweed cover increased dramatically, showing that *C. rodgersii* limited the distribution of seaweeds.

Figure 52.20

Effects of feeding by sea urchins on seaweed distribution.

Graph of percent seaweed cover versus time for sea urchins removed and control (sea urchins present).

Seaweed abundance in areas from which the long-spined sea urchin (Centrostephanus rodgersii) had been removed was much higher than seaweed abundance in adjacent control sites from which the urchin was not removed.

In addition to predation and herbivory, the presence or absence of pollinators, food resources, parasites, pathogens, and competing organisms can act as a biotic limitation on species distribution. Such biotic limitations are common in nature.

Abiotic Factors

The last question in the flowchart in Figure 52.18 considers whether abiotic factors, such as temperature, water, oxygen, salinity, sunlight, or soil, might be limiting a species' distribution. If the physical conditions at a site do not allow a species to survive and reproduce, then the species will not be found there. Throughout this discussion, keep in mind that most abiotic factors vary substantially over space and time. Daily and annual fluctuations of abiotic factors may either blur or accentuate regional distinctions. Furthermore, organisms can avoid some stressful conditions temporarily through behaviors such as dormancy or hibernation (see Concept 40.4).

Temperature

Environmental temperature is an important factor in the distribution of organisms because of its effect on biological processes. Cells may rupture if the water they contain freezes (at temperatures below 0°C), and the proteins of most organisms denature at temperatures above 45°C. Organisms typically function best within a specific range of environmental temperature. Temperatures outside that range may force some animals to expend energy regulating their internal temperature, as mammals and birds do (see Figure 40.16). Extraordinary adaptations enable certain organisms, such as thermophilic prokaryotes, to live outside the temperature range habitable by other life.

As mentioned earlier, climate change has already caused hundreds of species to alter their geographic ranges. A shift in the range of one species can also have profound effects on the distribution of other species. Consider how rising sea temperatures have affected the geographic range of the sea urchin *C. rodgersii*. Since 1950, water temperatures along the coast of Tasmania, an island south of mainland Australia, have increased from 11.5°C to 12.5°C. This has enabled *C. rodgersii*—whose larvae fail to develop properly if temperatures drop below 12°C—to expand its range to the south **(Figure 52.21)**. The urchin is a voracious consumer of kelp and other algae. As a result, algal communities that once harbored a rich diversity of other species have been completely destroyed in regions where the urchin has become well established (denoted by a solid orange line in Figure 52.21).

Figure 52.21

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A sea urchin’s expanding range.
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Map of Southeast Australia and the island of Tasmania, showing the expanding range of the sea urchin C. rodgerii.

Since the 1950s, water temperatures along the coast of Tasmania have increased, allowing the sea urchin C. rodgersii to expand its range to the south. The bold orange type indicates the years when C. rodgersii was first observed colonizing those locations. Once its population has been established in a new location, C. rodgersii has eliminated the local kelp community.

Map shows the southeastern coast of Australia and the island of Tasmania, which is south of Australia. The East Australian current flows south in the ocean about 600 km east of the Australian coast. Originally, the southeast coast of Australia was the southern portion of C. rodgerii’s range. In the 1960s, C. rodgerii’s range expanded southward to the islands between Australia and Tasmania. By 1978 the sea urchin’s range had expanded to the eastern coast of Tasmania. During the 1980s C. rodgerii expanded all the way down the eastern coast of Tasmania, and the kelp communities in this area were decimated. By 2005, C. rodgerii’s range included the southern coast of Tasmania, threatening the kelp communities there.

Water and Oxygen

The dramatic variation in water availability among habitats is another important factor in species distribution. Species living at the seashore or in tidal wetlands can desiccate (dry out) as the tide recedes. Terrestrial organisms face a nearly constant threat of desiccation, and the distribution of terrestrial species reflects their ability to obtain and conserve water. Many amphibians, such as the tiny frog in Figure 52.1, are particularly vulnerable to drying because they use their moist, delicate skin for gas exchange. Desert organisms exhibit a variety of adaptations for acquiring and conserving water in dry environments, as described in Concept 44.4.

Water affects oxygen availability in aquatic environments and in flooded soils, where the slow diffusion of oxygen in water can limit cellular respiration and other physiological processes. Oxygen concentrations can be particularly low in deep ocean and deep lake waters, as well as in sediments where organic matter is abundant. Flooded wetland soils may also have low oxygen content. Mangroves and other trees have specialized roots that project above the water and help the root system obtain oxygen (see Figure 35.4). Unlike many flooded wetlands, the surface waters of streams and rivers tend to be well oxygenated because of rapid exchange of gases with the atmosphere.

Salinity

The salt concentration of water in the environment affects the water balance of organisms through osmosis. Most aquatic organisms are restricted to either freshwater or saltwater habitats by their limited ability to osmoregulate (see Concept 44.1). Most terrestrial organisms can excrete excess salts from specialized glands or in feces or urine. However, the salt concentrations in some habitats (such as salt flats) are so high that few species of plants or animals can survive there **(Figure 52.22)**. In the **Scientific Skills Exercise**, you can interpret data from an experiment investigating the influence of salinity on plant distributions.

Figure 52.22

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    Salar de Uyuni, in Bolivia, the largest salt flat in the world.
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Birds stand in shallow water in the middle of a salt flat. Mountains are present in the background.

Aside from the flamingos that breed there annually, very few animals or plants inhabit the vast expanse of crusty white salt.

Scientific Skills Exercise

Making a Bar Graph and a Line Graph to Interpret Data

APScience Practices 3, 4, 6

How Do Salinity and Competition Affect the Distribution of Plants in an Estuary? Field observations show that *Spartina patens* (salt marsh hay) is a dominant plant in salt marshes and *Typha angustifolia* (cattail) is a dominant plant in freshwater marshes. In this exercise, you will graph and interpret data from an experiment that examined the influence of an abiotic factor, salinity, and a biotic factor, competition, on the growth of these two species.

Spartina patens has a hay- or grain-like quality while Typha angustifolia has an oblong tail. Both are in marshes.

Spartina patens (on the left); Typha angustifolia (on the right)

How the Experiment Was Done Researchers planted *S. patens* and *T. angustifolia* in salt marshes and freshwater marshes with and without neighboring plants. After two growing seasons 1.5years, they measured the biomass of each species in each treatment. The researchers also grew both species in a greenhouse at six salinity levels and measured the biomass at each level after eight weeks.

Data from the Field Experiment (averages of 16 replicate samples)

A table lists the average biomass of Spartina patens and Typha angustifolia with and without neighbors in salt and freshwater marshes.

The table is as follows:

With or without neighbors	Average biomass, Spartina patens. Salt marshes	Average biomass, Spartina patens. Freshwater marshes	Average biomass, Typha angustifolia. Salt marshes	Average biomass, Typha angustifolia. Freshwater marshes
With	8	3	0	18
Without	10	20	0	33

Average biomass is given in grams per 100 centimeters squared.

Data from the Greenhouse Experiment

A table lists salinity and percentage of maximum biomass of S. patens and T. angustifolia.

The table is as follows:

Salinity (parts per thousand)	0	20	40	60	80	100
Percent maximum biomass (S. patens)	77	40	29	17	9	0
Percent maximum biomass (T. angustifolia)	80	20	10	0	0	0

Data from C. M. Crain et al., Physical and biotic drivers of plant distribution across estuarine salinity gradients, *Ecology* 85:2539–2549 (2004).

INTERPRET THE DATA

- Make a bar graph of the data from the field experiment. (For additional information about graphs, see the Scientific Skills Review in Appendix D.) What do these data indicate about the salinity tolerances of *S. patens* and *T. angustifolia*?
- What do the data from the field experiment indicate about the effect of competition on the growth of these two species? Which species was limited more by competition?
- Make a line graph of the data from the greenhouse experiment. Decide which values constitute the dependent and independent variables, and use these values to set up the axes of your graph.
-
    - In the field, *S. patens* is typically absent from freshwater marshes. Based on the data, does this appear to be due to salinity or competition? Explain your answer.

- *T. angustifolia* does not grow in salt marshes. Does this appear to be due to salinity or competition? Explain your answer.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Salmon that migrate between freshwater streams and the ocean use both behavioral and physiological mechanisms to osmoregulate. They balance their salt content by adjusting the amount of water they drink and by switching their gills from taking up salt in fresh water to excreting salt in the ocean.

Sunlight

Sunlight provides the energy that drives most ecosystems, and too little sunlight can limit the distribution of photosynthetic species. In forests, shading by leaves makes competition for light especially intense, particularly for seedlings growing on the forest floor. In aquatic environments, every meter of water depth

absorbs about 45% of the red light and about 2% of the blue light passing through it. Thus, most photosynthesis occurs relatively near the water surface.

Too much light can also limit the survival of organisms. In some ecosystems, such as deserts, high light levels can increase temperature stress if animals and plants are unable to avoid the light or to cool themselves through evaporation (see Figure 40.12). At high elevations, the sun's rays are more likely to damage DNA and proteins because the atmosphere is thinner, absorbing less ultraviolet (UV) radiation. Damage from UV radiation, combined with other abiotic stresses, prevents trees from surviving above a certain elevation, resulting in the appearance of a tree line on mountain slopes **(Figure 52.23)**.

Figure 52.23

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Alpine tree line in Banff National Park, Canada.
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Vegetation stops growing at a particular altitude on a mountain range, above which snow is present.

Organisms living at high elevations are exposed not only to high levels of ultraviolet radiation but also to freezing temperatures, moisture deficits, and strong winds. Above the tree line, the combination of such factors restricts the growth and survival of trees.

Rocks and Soil

In terrestrial environments, the pH, mineral composition, and physical structure of rocks and soil limit the distribution of plants and thus of the animals that feed on them, contributing to the patchiness of terrestrial ecosystems. The pH of soil can limit the distribution of organisms directly, through extreme acidic or basic conditions, or indirectly, by affecting the solubility of toxins and nutrients. Soil phosphorus, for instance, is relatively insoluble in basic soils and precipitates into forms unavailable to plants.

In a river, the composition of rocks and soil that make up the substrate (riverbed) can affect water chemistry, which in turn influences the resident organisms. In freshwater and marine environments, the structure of the substrate determines the organisms that can attach to it or burrow into it.